

[Riboflavin: Fact Sheet for Consumers \(original version\)](#).

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What is riboflavin and what does it do?

Riboflavin (also called vitamin B2) is important for the growth, development, and function of the cells in your body. It also helps turn the food you eat into the energy you need.

How much riboflavin do I need?

The amount of riboflavin you need depends on your age and sex. Average daily recommended amounts are listed below in milligrams (mg).

Life Stage	Recommended Amount
Birth to 6 months	0.3 mg
Infants 7–12 months	0.4 mg
Children 1–3 years	0.5 mg
Children 4–8 years	0.6 mg
Children 9–13 years	0.9 mg
Teen boys 14–18 years	1.3 mg
Teen girls 14–18 years	1.0 mg
Men	1.3 mg
Women	1.1 mg
Pregnant teens and women	1.4 mg
Breastfeeding teens and women	1.6 mg

What foods provide riboflavin?

Riboflavin is found naturally in some foods and is added to many fortified foods. You can get recommended amounts of riboflavin by eating a variety of foods, including the following:

- Eggs, organ meats (such as kidneys and liver), lean meats, and low-fat milk

- Some vegetables (such as mushrooms and spinach)
- Fortified cereals, bread, and grain products

What kinds of riboflavin dietary supplements are available?

Riboflavin is found in multivitamin/mineral supplements, in B-complex dietary supplements, and in supplements containing only riboflavin. Some supplements have much more than the recommended amounts of riboflavin, but your body can't absorb more than about 27 mg at a time.

Am I getting enough riboflavin?

Most people in the United States get enough riboflavin from the foods they eat and deficiencies are very rare. However, certain groups of people are more likely than others to have trouble getting enough riboflavin:

- Athletes who are vegetarians (especially strict vegetarians who avoid dairy foods and eggs).
- Pregnant women and breastfeeding women and their babies.
- People who are vegan.
- People who do not eat dairy foods.
- People with a genetic disorder called riboflavin transporter deficiency. This disorder prevents the body from properly absorbing and using riboflavin, causing riboflavin deficiency.

What happens if I don't get enough riboflavin?

You can develop riboflavin deficiency if you don't get enough riboflavin in the foods you eat or if you have certain diseases or hormone disorders.

Riboflavin deficiency can cause skin disorders, sores at the corners of your mouth, swollen and cracked lips, hair loss, sore throat, liver disorders, and problems with your reproductive and nervous systems.

Severe, long-term riboflavin deficiency causes a shortage of red blood cells (anemia), which makes you feel weak and tired. It also causes clouding of the lens in your eyes (cataracts), which affects your vision.

What is an effect of riboflavin supplements on health?

Scientists are studying riboflavin to better understand how it affects health. Here is an example of what this research has shown.

Migraine headache

Some studies show that riboflavin supplements might help prevent migraine headaches, but other studies do not. Riboflavin supplements usually have very few side effects, so some medical experts recommend trying riboflavin, under the guidance of a health care provider, for preventing migraines.

Can riboflavin be harmful?

Riboflavin has not been shown to cause any harm.

Does riboflavin interact with medications or other dietary supplements?

Riboflavin is not known to interact with any medications. However, it's always important to tell your doctor, pharmacist, and other health care providers about any dietary supplements and prescription or over-the-

counter medicines you take. They can tell you if the dietary supplements might interact with your medicines or if the medicines might interfere with how your body absorbs, uses, or breaks down nutrients.

Riboflavin and healthful eating

People should get most of their nutrients from food and beverages, according to the federal government's *Dietary Guidelines for Americans*. Foods contain vitamins, minerals, dietary fiber, and other components that benefit health. In some cases, fortified foods and dietary supplements are useful when it is not possible to meet needs for one or more nutrients (for example, during specific life stages such as pregnancy). For more information about building a healthy dietary pattern, see the [Dietary Guidelines for Americans](#).

Where can I find out more about riboflavin?

- For general information on riboflavin
 - Office of Dietary Supplements (ODS) Health Professional Fact Sheet on [Riboflavin](#)
 - [Riboflavin](#), MedlinePlus
- For more information on food sources of riboflavin
 - ODS Health Professional Fact Sheet on [Riboflavin](#)
 - USDA's [FoodData Central](#)
 - Nutrient List for riboflavin listed by [food](#) or by [riboflavin content](#), USDA
- For more advice on choosing dietary supplements
 - ODS [Frequently Asked Questions: Which brand\(s\) of dietary supplements should I purchase?](#)
- For information about building a healthy dietary pattern
 - [Dietary Guidelines for Americans](#)

Glossary

absorption

In nutrition, the process of moving protein, carbohydrates, fats, and other nutrients from the digestive system into the bloodstream. Most absorption occurs in the small intestine.

adverse event

An unwanted medical problem that occurs during treatment. Adverse events may be unrelated to the treatment or they may be caused by the therapy or procedure. For example, an adverse event may be caused by the toxic effects of a particular drug or dietary supplement or by an interaction with another therapy. Also called adverse effect and side effect.

anemia

A condition in which the number of red blood cells in the blood, or the amount of hemoglobin in them, is lower than normal, causing a condition in which red blood cells are not able to supply enough oxygen to all the tissues in the body. Hemoglobin is the substance in red blood cells that carries oxygen to your body's cells.

cataract

A condition in which the lens of the eye becomes cloudy. Symptoms include blurred, cloudy, or double vision; sensitivity to light; and difficulty seeing at night. Without treatment, cataracts can cause blindness. Cataracts occur in people of all ages but are most common in the elderly.

cell

The individual unit that makes up the tissues of the body. All living things are made up of one or more cells, which are the smallest units of living structure capable of independent existence.

dairy food

Milk and products made with milk, such as buttermilk, yogurt, cheese, cottage cheese, and ice cream.

deficiency

An amount that is not enough; a shortage.

dietary fiber

A substance in plants that you cannot digest. It adds bulk to your diet to make you feel full, helps prevent constipation, and may help lower the risk of heart disease and diabetes. Good sources of

dietary fiber include whole grains (such as brown rice, oats, quinoa, bulgur, and popcorn), legumes (such as black beans, garbanzo beans, split peas, and lentils), nuts, seeds, fruit, and vegetables.

Dietary Guidelines for Americans

Advice from the federal government to promote health and reduce the chance (risk) of long-lasting (chronic) diseases through nutrition and physical activity. The Guidelines are updated and published every 5 years by the US Department of Health and Human Services and the US Department of Agriculture.

dietary supplement

A product that is intended to supplement the diet. A dietary supplement contains one or more dietary ingredients (including vitamins, minerals, herbs or other botanicals, amino acids, and other substances) or their components; is intended to be taken by mouth as a pill, capsule, tablet, or liquid; and is identified on the front label of the product as being a dietary supplement.

disorder

In medicine, a disturbance of normal functioning of the mind or body. Disorders may be caused by genetic factors, disease, or trauma.

fortified

When nutrients (such as vitamins and minerals) are added to a food product. For example, when calcium is added to orange juice, the orange juice is said to be "fortified with calcium". Similarly, many breakfast cereals are "fortified" with several vitamins and minerals.

genetic disorder

A disease or disorder caused by an alteration or variation (mutation) in a gene or group of genes in the cells of an individual. Examples of genetic disorders include breast cancer, cystic fibrosis, Parkinson's disease, and celiac disease. They can be inherited or can occur without a known cause.

hormone

A group of chemicals made by glands in the body. Hormones circulate in the bloodstream and control the actions of certain cells or organs. Some hormones can also be manufactured.

infant

A child younger than 12 months old.

interaction

A change in the way a dietary supplement acts in the body when taken with certain other supplements, medicines, or foods, or when taken with certain medical conditions. Interactions may cause the dietary supplement to be more or less effective, or cause effects on the body that are not expected.

kidney

One of two organs that remove waste from the blood (as urine). The kidneys also make erythropoietin (a substance that stimulates red blood cell production) and help regulate blood pressure. The kidneys are located near the back under the lower ribs.

lens

The clear part of the eye behind the iris that changes shape to focus near and far objects onto the retina (the nerve tissue at the back of the eye that receives images and sends them to the brain).

liver

A large organ located in the right upper abdomen. It stores nutrients that come from food, makes chemicals needed by the body, and breaks down some medicines and harmful substances so they can be removed from the body.

migraine

A type of headache that causes intense throbbing or pulsing pain, sensitivity to light and sound, and nausea and vomiting. A migraine often begins with visions of flashing lights, zigzag lines, and/or temporary loss of sight.

milligram

mg. A measure of weight. It is a metric unit of mass equal to 0.001 gram (it weighs 28,000 times less than an ounce).

mineral

In nutrition, an inorganic substance found in the earth that is required to maintain health.

nervous system

The brain and spinal cord, including the network of nerves that carry messages back and forth between the brain and all parts of the body. The nervous system controls what the body does.

nutrient

A chemical compound in food that is used by the body to function and maintain health. Examples of nutrients include proteins, fats, carbohydrates, vitamins, and minerals.

Office of Dietary Supplements

ODS, Office of Disease Prevention, Office of Director, National Institutes of Health, Department of Health and Human Services. ODS strengthens knowledge and understanding of dietary supplements by evaluating scientific information, stimulating and supporting research, disseminating research results, and educating the public to foster an enhanced quality of life and health for the US population.

pharmacist

A person licensed to make and dispense (give out) prescription drugs and who has been taught how they work, how to use them, and their side effects.

prescription

A written order from a health care provider for medicine, therapy, or tests.

prevent

To stop from happening.

red blood cell

A cell that carries oxygen to and removes carbon dioxide from all parts of the body.

riboflavin

A nutrient that is needed by the body for growth, cell function, and to make energy from food. It works together with other B vitamins, and acts as an antioxidant to protect cells from free radical damage. Riboflavin is found in some foods, including liver, mushrooms, spinach and leafy green vegetables, milk and dairy products, eggs, whole grains, enriched breads and cereals, and lean meats. Also called vitamin B₂.

supplement

A nutrient that may be added to the diet to increase the intake of that nutrient. Sometimes used to mean dietary supplement.

US Department of Agriculture

USDA promotes America's health through food and nutrition, and advances the science of nutrition by monitoring food and nutrient consumption and updating nutrient requirements and food composition data. USDA is responsible for food safety, improving nutrition and health by providing food assistance and nutrition education, expanding markets for agricultural products, managing and protecting US public and private lands, and providing financial programs to improve the economy and quality of rural American life.

vegan

A person who eats only plant-based foods. Vegans do not eat meat, poultry, fish, eggs, milk, dairy products, or honey, and do not use leather, silk, or wool, or soaps and cosmetics that are made from animal products.

vegetarianism

The practice of avoiding all or most animal products for environmental, philosophical, and health reasons. Vegetarians (people who practice vegetarianism) eat a diet based on foods that come from plants and may include some dairy products and eggs. See: vegetarian diet.

vitamin

A nutrient that the body needs in small amounts to function and maintain health. Examples are vitamins A, C, and E.

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